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Stress urinary incontinence: treatment

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Introduction

- If stress urinary incontinence leads to psychological or social distress, treatment is warranted.
- Conservative therapeutic modalities are usually the first step (first line therapy).
- A sufficient improvement in continence can be expected in daily routine in approx. 50 % to 80 % of cases with conservative treatment.
- Suburethral sling surgery is the gold standard in terms of surgical treatment for stress urinary incontinence. The success rate is approx. 85 %.
- If there is a concomitant downward displacement of the genital organs, or a relevant psychological suffering and a patients desire for surgery, a primary surgical treatment for anatomic correction can be appropriate.
- For a successful conservative treatment, the patient's active participation and thus motivation is paramount.
- The aim of urinary incontinence physiotherapy is to teach awareness (pelvic floor muscles, respiration, etc.), to strengthen the pelvic floor muscles, and to practice reflexive pelvic floor activity (e.g., when coughing or lifting a load). The following therapeutic modalities are currently available: Pelvic floor muscle training and electrical stimulation of the pelvic floor. Both can be combined with biofeedback. The choice between these treatment modalities is guided by whether or not the patient can correctly activate the pelvic floor. If testing (ability to contract the pelvic floor; see Diagnosis) shows that the patient can contract the pelvic floor autonomously, she will be instructed in pelvic floor muscle training. If no voluntary contraction can be felt, electrical stimulation and biofeedback are indicated.

Pelvic floor training

Independent pelvic floor muscle training: Women who can independently activate the pelvic floor muscles must train the pelvic floor for a few minutes, several times a day, by contracting the muscles, and then holding the contraction for 5 seconds before relaxing for 10 seconds. This improves muscle strength and the neuromuscular coordination. It is also essential that reflexive contraction on physical stress (coughing, sneezing etc.) is recognized and employed.

Not all women are able to train their pelvic floor muscles independently. Contrary to other muscle groups within the body, the pelvic floor muscles are not visible. There are ways to make patients “more aware” of the pelvic floor in order to facilitate conditioning and training success. Pelvic floor activity can be demonstrated using additional tools like biofeedback, providing the patient with feedback on the success of her muscle training (figure 1).

Prior to the start of pelvic floor training, information on the training sessions and homework as well as the chances of success must be given to the patient. For electrical stimulation and biofeedback it is essential that the patient understands that a probe is inserted into the vagina, and that roughly 50 to 60 % of women are satisfied with their success in the first year after training. If training is neglected, urinary incontinence often deteriorates again.

In most cases, conservative treatment does not enable complete continence to be achieved. However, the effect is often sufficient for women to cope with in everyday life and therefore be satisfied.

Figure 1: Physiotherapy with electrical stimulation and biofeedback. The patient can insert the vaginal probe herself. With biofeedback, the physiotherapist can instruct the patient on correct pelvic floor training using the EMG curve taken from the pelvic floor. In addition, the probe can emit a current pulse to trigger a muscle contraction.

Drugs/medical treatment

Estrogen is potentially an option, though it primarily improves the symptoms of vaginal atrophy in postmenopausal women, rather than stress urinary incontinence. They are therefore only an additive therapy. Local treatment with estrogen cream or ovules is sufficient. [B06][B07]

Estriol, e.g., Ovestin® Cream and Ovules or Oestro Gynaedron® Cream are applied vaginally once or twice per week in the evening. The creams are supplied with an acrylic tube. The tube is filled with the cream and must be inserted vaginally. The stamp is then pressed in order to introduce the correct amount of cream into the vagina.

In some EU countries, although not yet Switzerland, duloxetine (Yentreve®) has been authorized since fall 2004. Duloxetine is a reuptake inhibitor for serotonin and norepinephrine.

Duloxetine probably increases the neurotransmitter concentration and thus, the tone and contractility of the internal bladder sphincter. Studies show that the frequency of unintentional urination is reduced significantly with this drug in over 50 % of patients, and quality of life is improved. However, full recovery proves to be rare.
Yentreve® dosage = one 40mg tablet each morning and evening. A gradual increase in dose with 2 x 20 mg per day can reduce side effects like dizziness and nausea.

Pessary treatment

Various pessaries are available on the market; they are inserted vaginally and can lead to an improvement of the urethral closure pressure and thus, a reduction of stress urinary incontinence. Pessary treatment can provide a valuable improvement in stress urinary incontinence, particularly in patients with contraindications for surgical procedures. Such an approach can be successful with little effort in women who are only incontinent during exercise. Pessary treatment can therefore be seen as a palliative option, which helps to avoid or to delay an operation. Urethra pessaries (figure 2) or so-called incontinence tampons are mainly used.

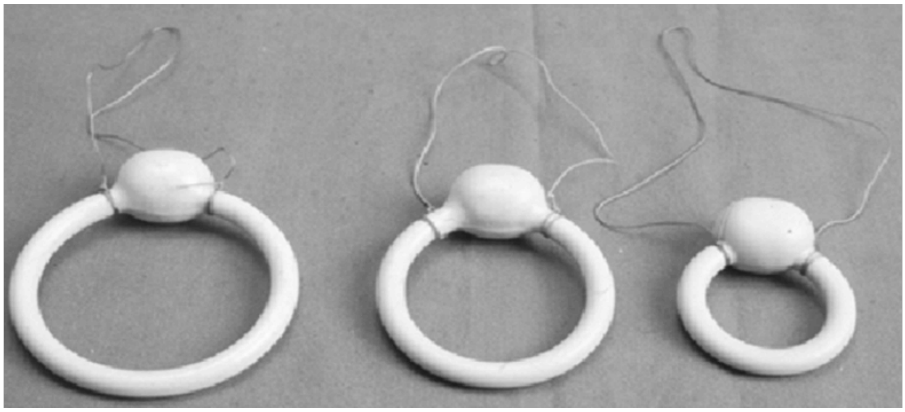

Figure 2: Urethra pessary of varying sizes.

Incontinence surgery

If the symptoms of stress urinary incontinence cannot be sufficiently treated with conservative treatment options, incontinence surgery is indicated.

Today, suburethral sling surgery is the “gold standard”. TVT (tension-free vaginal tape) was the first introduced sling and is currently the most well-known sling procedure. TVT and subsequent sling procedures provide minimally invasive options for vaginal continence surgery, showing very good results for continence with minimal peri- and post-operative morbidity.

Sling surgery has largely replaced the previous colposuspension techniques. To date, the success rate of TVT is at least equivalent to the open abdominal Burch colposuspension; however, the cost of the surgery is lower, with post-operative shorter hospitalization time. A polypropylene is applied to create a support under the mid urethra (figure 3). Following the surgery, in situations where the intrabdominal pressure is increased (coughing, pressing, exercise), the urethra is then compressed against the TVT tape and supports the continence mechanism.

Suburethral sling surgery may be associated with complications, such as bladder injury, hemorrhage, infection, and wound healing disorders. Severe complications, such as injury of the large blood vessels, bowel or nerves, prove to be extremely rare. In addition to the proper training of surgeons, it is paramount that women are carefully informed of possible complications so that the expected effect can be weighed against the potential side effects.

The major success of TVT surgery has led to a variety of new sling techniques coming onto the market. Most of these methods are closely related to TVT. The largest difference is in the surgical technique due to a different insertion approach (transobturator) or shorter slings (minislings). The tape material used as today's standard is a woven monofilament polypropylene mesh for both procedures.

The most common intraoperative complication of TVT insertion is a bladder injury, which is why retropubic slings must always be combined with cystoscopy. When detected during surgery by cystoscopy and handled correctly, this issue can be corrected with no further disadvantage to the patient. Nevertheless, an approach that avoids the retropubic area has been developed. Hereby the tape is inserted through the obturator foramen (TOT, TVT-O) (figure 4).

Comparative studies showed that the transobturator procedure is equivalent to the retropubic approach in terms of success rate, but that in addition to the reduced risk of bladder injury, other, new issues can occur, such as inguinofemoral pain for several days post-surgery, and sometimes beyond, lateral vaginal mesh erosion with corresponding cohabitation issues can also be seen. Here, in-depth training and compliance with the standard operating steps are essential to avoid unnecessary complications. The cost-benefit-risk analysis reveals a better overall outcome for retropubic slings.

According to current data, laser therapy has been able to achieve a slight improvement in symptoms of mild stress incontinence [P10].

Another form of surgical therapy is the injection of bulking agents under the urethral mucosa.
The advantage of this method is that it can be performed in LA and has very few complications. However, the long-term results are not comparable with sling operations [P11].

The Swiss Working Group for Urogynecology and Pelvic Floor Pathology (AUG, Swiss Working Group for Gynecology), together with the Specialist Society for Urology, the Swiss Medical Association and quality experts, has published an Expert Opinion with recommendations on sling surgery. We recommend taking this Expert Opinion into account and acting accordingly [P09]. See also expert letter Nr. 44 (about sling operations for the treatment of female stress incontinence (stress incontinence)) of Swiss Society for Gynecology and Obstetrics (SGGG) (available in German and

French). [P09][P10][P11]

gynécologie Suisse SGGG – expert letter

In addition, the AUG, together with the AGUB (German society) and the AUB (Austria), was involved in the creation of a joint guideline that comprehensively covers all topics relating to female incontinence [P10]. [P10]

In experienced hands, sling surgery is very effective (objective recovery rate approx. 85 %). Potential risks include:

- Bladder perforation
- Pain (generally short term, rarely long term)
- Dyspareunia (painful intercourse)
- Incision site healing disorders (erosion)
- Blood vessel/bowel damage

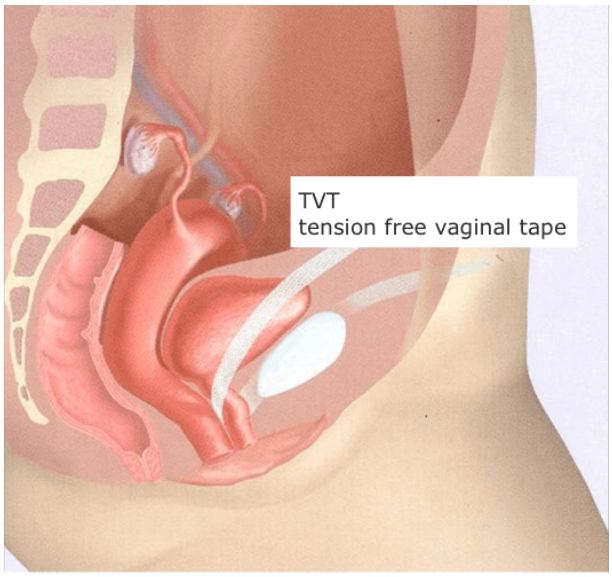

Figure 3: TVT (tension free vaginal tape) inserted behind the pubic symphysis and under the mid-section of the urethra.

The plastic tape is used to establish a pubo-urethral anchor point for the mid-third of the urethra:

- The TVT should neither elevate nor obstruct the urethra.
- It should act as a support to improve pressure transmission in case of physical stress.

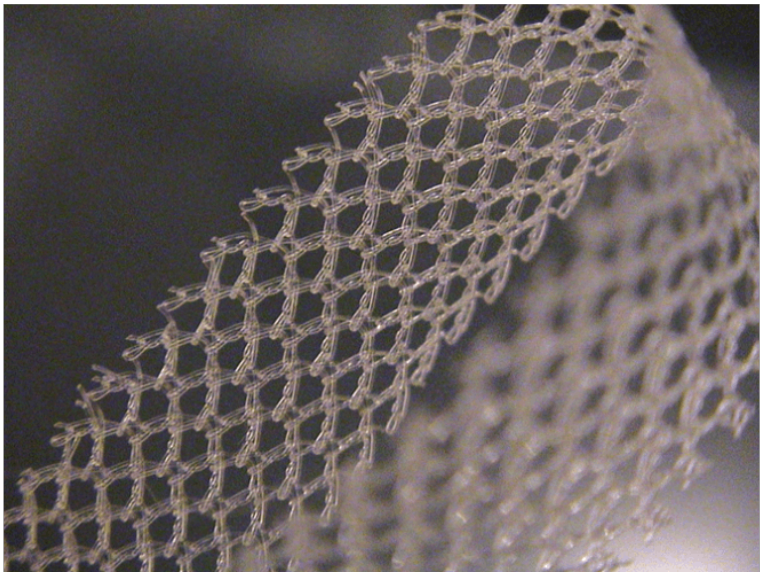

Figure 4: Polypropylene tape.

Operation

The operation can be performed under local anesthetic. First, a small incision is made in the vaginal skin under the mid-section of the urethra, and a narrow tunnel is formed to the pubic bone using scissors. The TVT sling has a sharp guide needle at each end, which can be used to push the mesh through the tissue. The polypropylene tape has a plastic covering enabling it to slide through the tissues more easily. A handle is employed to clamp the needle, with the tape lifted retropubically on either side of the urethra, then placed loosely under the urethra's mid-section. While the patient coughs repeatedly, the tape is tightened until almost no urine flows out. Care must be taken to ensure that, as a whole, the tape is free of tension.

The correct placement of the band is crucial to the success of the operation:

- if it is too loose, the desired continence will not be achieved.
- if it is too tight, this may lead to voiding disorders with increased residual urine volume and, in some cases, symptoms of an overactive bladder.

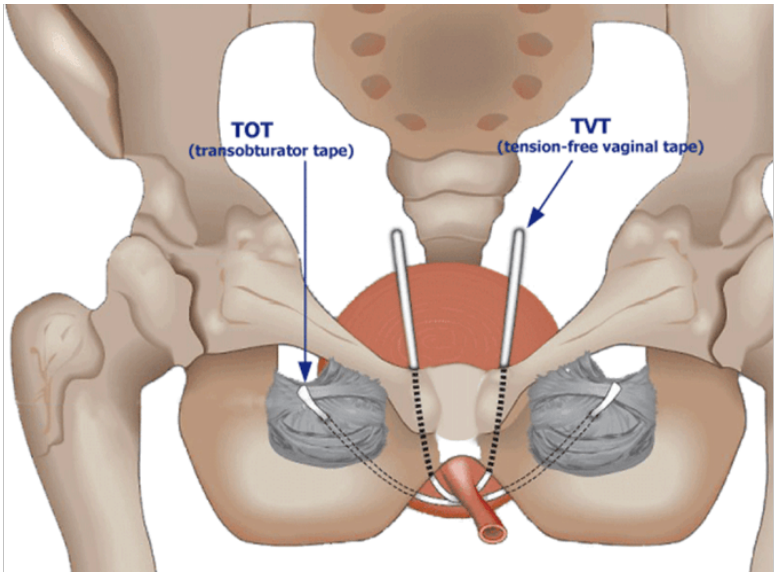

Figure 5: Comparison TVT/TOT

Literature

Nr.	Title	Author	Journal	Volume	Pages	Yea
[B01]	Pelvic floor muscle training is effective in treatment of female stress urinary incontinence, but how does it work?	Bo K	Int Urogynecol J Pelvic Floor Dysfunct	15(2)	76-84	200
[B02]	Transobturator and retropubic tape procedures in stress urinary incontinence: A systematic review and meta-analysis of effectiveness and complications	Latthe PM, Foon R, Toozs-Hobson P	BJOG	114	522-531	200
[B03]	Retropubic versus transobturator midurethral slings for stress incontinence	Richter HE, Albo ME, Zyczynski HM, Kenton K, Norton PA, Sirls LT, et al	NEJM	362	2066-2076	201
[B04]	Seventeen years' follow-up of the tension-free vaginal tape procedure for female stress urinary incontinence	Nilsson CG, Palva K, Aarnio R, Morcos E, Falconer C	Int Urogynecol J	24(8)	1265-1269	201

Nr.	Title	Author	Journal	Volume	Pages	Yea
[B05]	Minimally invasive synthetic suburethral sling operations for stress urinary incontinence in women: a short version Cochrane review	Ogah J, Cody DJ, Rogerson L	Neurourol Urodyn	30(3)	284-291	201
[B06]	The Effect of Vaginal Oestriol Cream on Subjective and Objective Symptoms of Stress Urinary Incontinence and Vaginal Atrophy: An International Multi-Centre Pilot Study	Weber M A, Lim V, Oryszczyn J, Te West N, Souget J, Jeffery S, Roovers J P W R, Moore K H	Observational Study Gynecol Obstet Invest	82(1)	15-21	201
[B07]	The effect of 12 weeks of estriol cream on stress urinary incontinence post-menopause: A prospective multinational observational study	Te West N I D, Harris K, Jeffrey S, de Nie I, Parkin K, Roovers J-P, Moore K H	Observational Study Neurourol Urodyn	42(4)	799-806	202
[B08]	S2k-Leitlinie Harninkontinenz der Frau (Registernummer 015 - 091)	Deutsche Gesellschaft für Gynäkologie und Geburtshilfe e.V. (DGGG)	AWMF			202

Nr.	Title	Author	Journal	Volume	Pages
[P01]	Adherence to pelvic floor muscle exercise and long-term effect on stress urinary incontinence. A five-year follow-up study	Bo K	Scand J Med Sci Sports	5(1)	36-39
[P02]	A randomized controlled trial of duloxetine alone, pelvic floor muscle training alone, combined treatment and no active treatment in women with stress urinary incontinence	Ghoniem GM, Van Leeuwen JS, Elser DM, Freeman RM, Zhao YD, Yalcin I et al.	J Urol	173(5)	1647-1653
[P03]	Efficacy of pelvic floor muscle exercises in women with stress, urge, and mixed urinary incontinence	Nygaard IE, Kreder KJ, Lepic MM, Fountain KA, Rhomberg AT	American journal of obstetrics and gynecology	174	120-125
[P04]	Are slings now the gold standard treatment for the management of female urinary stress incontinence and if so which technique?	Bemelmans BL, Chapple CR	Curr Opin Urol	13	301-307
[P05]	Retropubic compared with transobturator tape placement in treatment of urinary incontinence: a randomized controlled trial	Laurikainen E, Valpas A, Kivela A et al.	Obstet Gynecol	109	4-11

Nr.	Title	Author	Journal	Volume	Pages
[P06]	A cost-utility analysis of tension-free vaginal tape versus colposuspension for primary urodynamic stress incontinence	Manca A, Sculpher MJ, Ward K, Hilton P	BJOG	110	255-262
[P07]	Adverse events over two years after retropubic or transobturator midurethral sling surgery: findings from the Trial of Midurethral Slings (TOMUS) study	Brubaker L, Norton PA, Albo ME, Chai TC, Dandreo KJ, Lloyd KL et al.	American journal of obstetrics and gynecology	205(5)	498 e1-6
[P08]	Retropubic versus transobturator midurethral slings for stress incontinence	Richter HE, Albo ME, Zyczynski HM, Kenton K, Norton PA, Sirls LT et al.	The New England journal of medicine	362(22)	2066-2076
[P09]	Expertenbrief Nr. 44 Schlingenoperation zur Behandlung der weiblichen Belastungsinkontinenz	gynécologie suisse SGGG			
[P10]	Vaginal Laser Therapy for Stress Urinary Incontinence: A Systematic Review of Prospective Randomized Clinical Trials	Ranjbar A, Mehrnoush V, Darsareh F, Kotb A, Zakaria A, Shekari M, Shirzadfard Jahromi M	Review J Menopausal Med	28(3)	103-111

Nr.	Title	Author	Journal	Volume	Pages
[P11]	Polyacrylamide hydrogel (Bulkamid®) for stress urinary incontinence in women: a systematic review of the literature	Devi Kasi A, Pergialiotis V, Perrea D N, Khunda A, Doumouchtsis S K	Int Urogynecol J	27(3)	367-375